

CALRIG · DESIGN DECISIONS REGISTER

CalRig design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Approved follow-ups carried out: value engineering restated (USD 448.50 on 20 lines, USD 36.50 over the target); items 8 and 9 to confirm (status light, door border); open decision 1 (badge placement)
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for all seven open decisions on 2026-10-02 (CLR-DDR-003 accepted); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open items gathered from the review note, the decision records and the build work

CalRig design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	To decide	Options	Recommendation	What it affects in the build	Source
1	Where the name plate and status light sit. The front frame covers the jacket's front edges to 6 mm below the top, so the design places them on a printed badge on the top panel, its lip on the frame's top edge; this raises the overall height from 371 to 383 mm	(a) badge on the top panel, as modelled; (b) badge recessed into a notch in the top panel's front edge, no added height, harder foam cut	(a): simplest to make, and the rig stays 17 mm under the 400 mm limit of R13	Front badge (section 3.15 and step 12 of the build plan), overall height	REVIEW 2026-10-02, approved follow-ups

All earlier open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The heat pump's sink bases are at least 120 x 110 mm and take four M4 clamp screws at 57 mm either side and 42 mm above and below centre	The bases clamp the wall over the 90 x 100 mm opening; the shell's clamp holes are cut to match	CLR-DDR-003, P3
2	An inner fin block and fan that reach about 0.20 K/W	R2 at 20 °C and 85 % RH depends on it	CLR-DDR-002, D7; CLR-PRC-001, open questions
3	The draw latches fit a 41 x 40 mm tab and reach a keeper 15 mm in from the tab edge	Sets the frame tab size and the keeper position	CLR-DDR-003, P1
4	The thread sizes of the bulkhead fittings (12 and 6 mm), cable glands (M20) and aerosol bulkhead (16 mm), and that the multi-hole gland inserts suit the sensor leads	The shell holes are laser cut to these sizes before anything is fitted	CLR-DDR-003, P7 to P11
5	The HEPA unit's ports are 12 mm and 80 mm apart at 117 mm above the bench, or can be drilled to that	The back-wall fittings run straight into them	CLR-DDR-003, P8
6	The SHT45 tolerance the supplier states (0.1 or 0.2 °C)	R5 is at risk if it is 0.2 °C	CLR-CAL-001, [F1]
7	The bimetal switches and relay: opening temperatures of 50 and 70 °C and a relay rated 10 A at 12 V direct current	R16 rests on them	CLR-DDR-003, Table 2
8	The 12 V panel indicator light is 5 mm with an 8 mm bezel and a built-in resistor, and fits a 5.2 mm hole in an 18 mm badge face	Sets the badge hole and the controller output	CLR-DEC-001, 2026-10-02 (appearance item 3)
9	The print shop can cut the 22 mm vinyl border with the four keeper notches, in a vinyl that takes the 40 °C, 85 % RH point on the door's outer face	The border must stay flat on the door	CLR-DEC-001, 2026-10-02 (appearance item 4)

■ Value engineering

Value-engineering target: USD 412 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 448.50 (USD 36.50 over the target).

Main cost drivers (CLR-CAL-001, K): the reference cluster (line 8, \$132), the chamber shell (line 1, \$51), the Peltier heat pump (line 5, \$40), the parts added for construction (\$27, of which new line 19, bulkhead fittings, cable glands and drain, is \$14) and the details adopted on 2026-10-02 (\$9.50: front badge, status light, name plate and lead, line 20, \$5.50; door border \$3; pull handle \$1). The CO₂ path (SCD30 in the reference cluster and soda lime, line 16, \$8) is the largest block that is not core to the temperature, humidity and particle functions.

Savings worth trying:

- A core version without CO₂ (no SCD30, no soda lime), about USD 381.50; this changes what the rig does and would be proposed, not made (CLR-DDR-003, A1).
- Re-pricing the fittings, glands and drain (line 19) and the front frame, drip tray and spacers (\$6) at purchase.
- Looking for savings in the reference cluster and shell; no single line obviously offers \$36.50.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items A1 to A6: budget raised to \$400, transfer references checked at salt and ice fixed points, accept the cooling limit (R1 redefined), incense smoke decay, carbon dioxide optional and nitrogen dioxide by field collocation only, insulated door panel	Amish: "i accept all your recommendations, go with them across all repos."	CLR-DDR-001, CLR-DDR-002
2026-09-25	Larger inner fin block (about 0.20 K/W); R13 relaxed to 14 kg; certified temperature probe only if the budget allows	Amish, same instruction	CLR-DDR-002, D7 to D9
2026-09-26	Budget set to \$412 to cover the priced BOM as it then stood	Amish: "i approve all the budget items."	CLR-DDR-002, O2
2026-10-02	Design for construction accepted: the changes P1 to P12 and their knock-on changes, including the 9 mm sealed birch plywood base, as made	Amish: "i approve your recommendations for all 555 open decisions."	CLR-DDR-003, Tables 1 and 2
2026-10-02	Mass margin of 0.17 kg accepted on paper; the finished rig is weighed at TRL 4, and a 5 mm acrylic top is the first fallback if it weighs over 14 kg	Amish: "i approve your recommendations for all 555 open decisions."	CLR-DDR-003, A2
2026-10-02	Base plate: sealed 9 mm birch plywood, as modelled; the renders are to show plywood too, so they match what will be built	Amish: "i approve your recommendations for all 555 open decisions."	CLR-DDR-003, A3; REVIEW 2026-09-26, item 2
2026-10-02	Collocation partner for the transfer SPS30: a state or local regulatory monitoring site with a federal equivalent PM2.5 monitor. First candidate to approach: a Texas Commission on Environmental Quality site in Dallas-Fort Worth; fallback: South Coast AQMD's AQ-SPEC program	Amish: "i approve your recommendations for all 555 open decisions."	CLR-DDR-001, O1
2026-10-02	Appearance model items: jacket facing (1) in renders only, decided at TRL 4; status light and name plate (3), printed door border (4, with no gland in the door, since P9 put the glands in the back wall) and door panel pull handle (5) adopted in the design; bubbler and dryer details, fan guards, example heads and valve lever (6 to 9) accepted as drawn	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, items 1 and 3 to 9
2026-10-02	Calibration report based on the US EPA 2021 performance targets and testing protocol for PM2.5 air sensors, with a stated uncertainty chain from the salt and ice fixed points and the collocation record	Amish: "i approve your recommendations for all 555 open decisions."	CLR-PRC-001, open questions
2026-10-02	R10 gains a large item case: a head up to 192 x 152 mm in plan takes a 2 x 2 block of four bays, and a head up to 192 x 70 mm takes two bays side by side, within the 244 mm clear height	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-25, cross-repo notes